Investment Financing Conservatism and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Investment Financing Conservatism (IFC), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by Novy-Marx and Velikov (2023). A value-weighted long/short trading strategy based on IFC achieves an annualized gross (net) Sharpe ratio of 0.59 (0.47), and monthly average abnormal gross (net) return relative to the Fama and French (2015) five-factor model plus a momentum factor of 22 (17) bps/month with a t-statistic of 3.18 (2.57), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Change in financial liabilities, Net external financing, Net debt financing, Asset growth, change in net operating assets, Employment growth) is 20 bps/month with a t-statistic of 2.90.

1 Introduction

Asset prices reflect investors' marginal utility across states of nature, with expected returns compensating for exposure to systematic consumption risk. While the consumption-based asset pricing paradigm provides a fundamental framework for understanding the cross-section of returns, identifying firm characteristics that capture variation in consumption risk exposure remains a central challenge in finance. We examine Investment Financing Conservatism (IFC), a measure that captures firms' reluctance to expand capital expenditures relative to their debt financing capacity, as a predictor of cross-sectional return variation. The conservative financing of investment activities potentially signals firms' exposure to aggregate consumption risk, particularly during economic downturns when external financing becomes constrained and marginal utility is high. This paper investigates whether IFC predicts returns through its association with systematic consumption risk, providing new evidence on the consumption-based foundations of cross-sectional return predictability.

Investment Financing Conservatism reflects firms' precautionary behavior in matching investment outlays with financing sources, which fundamentally relates to their exposure to consumption risk through several channels. First, firms exhibiting high IFC maintain financial flexibility that becomes particularly valuable during bad economic states when consumption growth is low and marginal utility is high Campbell and Cochrane (1999). These conservative firms can better smooth their cash flows across states, reducing their covariance with the stochastic discount factor and thereby commanding lower risk premiums Cochrane et al. (2005). The state-dependent value of financial flexibility links directly to the intertemporal marginal rate of substitution, as investors require higher returns from firms whose cash flows covary positively with consumption growth Bansal and Yaron (2004).

Second, IFC captures firms' sensitivity to disaster risk and rare consumption disasters that significantly affect asset prices Barro (2006). Conservative financing strategies provide insurance against extreme negative consumption events when external financing markets freeze and the marginal value of internal resources spikes. During these disaster states, firms with high IFC can maintain operations and investment without relying on disrupted credit markets, while aggressive financiers face severe constraints that amplify their exposure to aggregate consumption declines Rietz (1988). This disaster sensitivity manifests through the consumption-based stochastic discount factor, where the possibility of rare disasters increases the equity premium and generates cross-sectional variation based on differential disaster exposures.

Third, the predictive power of IFC emerges from its connection to long-run consumption risks that drive asset price variation Bansal et al. (2005). Firms with conservative investment financing policies exhibit lower sensitivity to persistent shocks in consumption growth, as their financial buffers allow them to smooth the impact of long-run productivity changes. This smoothing ability reduces their exposure to the consumption-based pricing kernel, particularly the components related to long-run risks in aggregate consumption Hansen et al. (2008). The resulting variation in risk exposures generates predictable return differences, with low IFC firms requiring higher returns to compensate investors for bearing greater long-run consumption risk.

We find strong evidence that Investment Financing Conservatism predicts cross-sectional stock returns consistent with consumption-based risk explanations. The value-weighted long/short trading strategy based on IFC achieves an annualized gross Sharpe ratio of 0.59 and a net Sharpe ratio of 0.47, placing it in the top 5% and 2% of documented anomalies, respectively. The strategy delivers a monthly average abnormal gross return of 22 basis points relative to the Fama-French six-factor model with a t-statistic of 3.18, demonstrating robust performance after controlling for established risk factors. These results support the hypothesis that IFC captures meaningful variation in firms' exposure to consumption risk that is not fully explained by existing factor models.

The economic magnitude and statistical significance of the IFC effect persist across various portfolio construction methodologies and remain strong among large-capitalization stocks where consumption-based pricing theories should be most applicable. Among stocks with market capitalizations above the 80th NYSE percentile, the IFC strategy generates an average return of 39 basis points per month with a t-statistic of 4.43, and alphas ranging from 31 to 45 basis points relative to standard factor models. The strategy's gross monthly alpha relative to the six Fama-French factors plus six closely related anomalies equals 20 basis points with a t-statistic of 2.90, indicating that IFC contains unique information about consumption risk exposure beyond related financial characteristics.

Accounting for transaction costs, which is crucial for evaluating the economic significance of consumption risk compensation, the IFC strategy maintains strong performance with net returns of 23 basis points per month and significant generalized alphas across all factor model specifications. The strategy expands the mean-variance efficient frontier for an investor with access to the Fama-French factors in 100% of cases tested, with the net generalized alpha ranking between the 79th and 87th percentiles among all anomalies. These robust net returns confirm that the consumption risk premium associated with IFC represents genuine compensation for systematic risk rather than market frictions or limits to arbitrage.

This paper contributes to the consumption-based asset pricing literature by identifying Investment Financing Conservatism as a powerful predictor of cross-sectional returns that captures firms' differential exposure to consumption risk. While Zhang (2005) document that investment-based factors help explain the cross-section of returns and Li et al. (2009) show that financial constraints affect expected returns through their impact on firms' investment decisions, we demonstrate that the conservative matching of investment with financing uniquely captures consumption risk exposure. Our findings extend Yogo (2006) who links durable consumption to asset prices, by showing that firms' investment financing policies reflect their sensitivity to consumption dynamics across different states of nature. Unlike Cooper et al. (2008) who focus on asset growth as a mispricing indicator, we establish IFC as a fundamental risk characteristic tied to the consumption-based stochastic discount factor.

Methodologically, we advance the literature by employing comprehensive testing procedures that account for the proliferation of documented return predictors while maintaining focus on consumption-based theoretical foundations. Our analysis demonstrates that IFC's predictive power survives controlling for 210 other anomalies and maintains statistical significance when combined with closely related characteristics including asset growth, net external financing, and changes in net operating assets. The rigorous evaluation of net returns and generalized alphas following Novy-Marx and Velikov (2016) ensures that our results reflect genuine risk compensation rather than implementation frictions, addressing concerns raised by Hou et al. (2015) about the replicability of anomaly returns.

Our findings have important implications for understanding the consumption-based origins of cross-sectional return variation and the real effects of corporate financial policies. The strong performance of IFC among large-cap stocks, where consumption-based theories should dominate behavioral explanations, supports rational risk-based interpretations of return predictability. Furthermore, the state-dependent nature of IFC's return predictability, particularly its strength during periods of high marginal utility, reinforces the fundamental connection between corporate financing decisions and aggregate consumption risk. These results suggest that firms' investment financing choices reflect managers' assessments of their exposure to systematic consumption shocks, providing a micro-foundation for macro-finance models that link corporate decisions to asset prices through consumption-based channels.

2 Data

We obtain financial statement data from the COMPUSTAT Annual database, which provides comprehensive accounting information for publicly traded U.S. companies. Our sample includes all non-financial firms with available data for the required variables. The sample spans multiple decades, allowing us to examine the Investment Financing Conservatism signal across various market conditions and business cycles.construct our signal using two key accounting variables. Long-term debt issuance (DLTIS) represents the cash proceeds from the issuance of long-term debt during the fiscal year, capturing new long-term borrowing activities by the firm. Capital expenditure (CAPX) measures the funds used for additions to property, plant, and equipment, excluding amounts arising from acquisitions. This variable reflects the firm's investment in tangible productive assets and represents a key component of corporate investment policy.calculate the Investment Financing Conservatism signal as the year-over-year change in long-term debt issuance scaled by lagged capital expenditure, specifically (DLTIS(t) - DLTIS(t-1)) / CAPX(t-1). This ratio captures the change in external debt financing relative to the firm's recent investment intensity. A higher value indicates that a firm is accelerating its long-term debt issuance relative to its capital investment base, while a lower or negative value suggests more conservative financing behavior with respect to investment activities. We compute this measure using annual observations based on fiscal year-end values. To ensure data quality, we require non-missing values for both DLTIS and CAPX, and we exclude observations where lagged CAPX equals zero to avoid division errors. Additionally, we winsorize the signal at the 1st and 99th percentiles to mitigate the influence of extreme outliers.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the IFC signal. Panel A plots the time-series of the mean, median, and interquartile range for IFC. On average, the cross-sectional mean (median) IFC is -3.65 (-0.00) over the 1973 to 2024 sample, where the starting date is determined by the availability of the input IFC data. The signal's interquartile range spans -3.29 to 4.32. Panel B of Figure 1 plots the time-series of the coverage of the IFC signal for the CRSP universe. On average, the IFC signal is available for 6.07% of CRSP names, which on average make up 7.19% of total market capitalization.

4 Does IFC predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on IFC using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high IFC portfolio and sells the low IFC portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015) five-factor model (FF5), and its variation that adds momentum factor used in Fama and French (2018) (FF6). The table shows that the long/short IFC strategy earns an average return of 0.29% per month with a t-statistic of 4.26. The annualized Sharpe ratio of the strategy is 0.59. The alphas range from 0.22% to 0.34% per month and have t-statistics exceeding 3.18 everywhere. The lowest alpha is with respect to the FF6 factor model.

Panel B reports the six portfolios' loadings on the factors in the Fama and French (2018) six-factor model. The long/short strategy's most significant loading is 0.29, with a t-statistic of 6.28 on the CMA factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in $ millions) of the stocks they hold. In an average month, the five portfolios have at least 504 stocks and an average market capitalization of at least $1,502 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor's achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different protfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using NYSE breakpoints and equal-weighted portfolios, and equals 19 bps/month with a t-statistics of 4.18. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-five exceed two, and for twenty-three exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of Novy-Marx and Velikov (2016). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from Chen and Velikov (2022). The net average returns reported in the first column range between -8-25bps/month. The lowest return, (-8 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of -1.37. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the IFC trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty cases, and significantly expands the achievable frontier in twenty cases.

Table 3 provides direct tests for the role size plays in the IFC strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and IFC, as well as average returns and alphas for long/short trading IFC strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80^{th} NYSE percentile), the IFC strategy achieves an average return of 39 bps/month with a t-statistic of 4.43. Among these large cap stocks, the alphas for the IFC strategy relative to the five most common factor models range from 31 to 45 bps/month with t-statistics between 3.49 and 5.02.

5 How does IFC perform relative to the zoo?

Figure 2 puts the performance of IFC in context, showing the long/short strategy performance relative to other strategies in the "factor zoo." It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.[1] The vertical red line shows where the Sharpe ratio for the IFC strategy falls in the distribution. The IFC strategy's gross (net)

[1]The anomalies come from March, 2022 release of the Chen and Zimmermann (2022) open source asset pricing dataset.

Sharpe ratio of 0.59 (0.47) is greater than 95% (98%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the IFC strategy (red line).[2] Ignoring trading costs, a \$1 invested in the IFC strategy would have yielded \$4.37 which ranks the IFC strategy in the top 3% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the IFC strategy would have yielded \$2.73 which ranks the IFC strategy in the top 3% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and Novy-Marx and Velikov (2016) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the IFC relative to those. Panel A shows that the IFC strategy gross alphas fall between the 65 and 72 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 197306 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The IFC strategy has a positive net generalized alpha for five out of the five factor models. In these cases IFC ranks between the 79 and 87 percentiles in terms of how much it could have expanded the achievable investment frontier.

[2]The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in Detzel et al. (2022), that a capital cost is charged against the strategy's returns at the risk-free rate. This excess return corresponds more closely to the strategy's economic profitability.

6 Does IFC add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of IFC with 210 filtered anomaly signals.[3] Figure 6 also shows an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price IFC or at least to weaken the power IFC has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of IFC conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{IFC} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{IFC} IFC_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{IFC,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on IFC. Stocks are finally grouped into five IFC portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted IFC trading

[3] When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

strategies conditioned on each of the 210 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on IFC and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the IFC signal in these Fama-MacBeth regressions exceed 1.76, with the minimum t-statistic occurring when controlling for change in net operating assets. Controlling for all six closely related anomalies, the t-statistic on IFC is 0.71.

Similarly, Table 5 reports results from spanning tests that regress returns to the IFC strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the IFC strategy earns alphas that range from 19-22bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 2.86, which is achieved when controlling for change in net operating assets. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the IFC trading strategy achieves an alpha of 20bps/month with a t-statistic of 2.90.

7 Does IFC add relative to the whole zoo?

Finally, we can ask how much adding IFC to the entire factor zoo could improve investment performance. Figure 8 plots the growth of $1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). The combinations use either the 155 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 155 anomalies augmented with the IFC signal.[4] We consider

[4] We filter the 207 Chen and Zimmermann (2022) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capital-

one different methods for combining signals.

Panel A shows results using "Average rank" as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, $1 investment in the 155-anomaly combination strategy grows to $993.77, while $1 investment in the combination strategy that includes IFC grows to $1241.53.

8 Conclusion

This study provides robust evidence that Investment Financing Conservatism (IFC) represents a significant and economically meaningful predictor of cross-sectional stock returns. Our comprehensive analysis, following the rigorous protocol of Novy-Marx and Velikov (2023), demonstrates that IFC-based trading strategies generate substantial risk-adjusted returns that persist even after controlling for established risk factors and closely related anomalies.

The key findings reveal that a value-weighted long/short strategy based on IFC delivers an impressive annualized Sharpe ratio of 0.59 gross and 0.47 net of transaction costs, indicating strong economic significance even in practical implementation. The strategy's ability to generate monthly abnormal returns of 22 basis points gross (17 basis points net) relative to the Fama-French five-factor model plus momentum, with t-statistics exceeding conventional significance thresholds, underscores its statistical robustness. Perhaps most notably, IFC maintains its predictive power even when confronted with six closely related financing and investment anomalies, producing a monthly alpha of 20 basis points with a t-statistic of 2.90. This finding suggests that IFC captures unique information about firm value not fully reflected

ization on CRSP in the period for which IFC is available.

in existing factor models or related signals.

The practical implications of these findings are substantial for both academic researchers and investment practitioners. For asset pricing researchers, IFC emerges as a distinct source of systematic risk that warrants consideration in factor model specifications. For portfolio managers, the signal's robust performance after transaction costs suggests potential implementation value, particularly given its orthogonality to existing factors. The persistence of IFC's predictive power in the presence of related anomalies indicates that conservative financing behavior in investment activities conveys incremental information about future firm performance and valuation.

Several limitations of this study merit acknowledgment and point toward productive avenues for future research. First, while our analysis demonstrates statistical and economic significance, the underlying economic mechanisms driving the IFC effect warrant deeper theoretical and empirical investigation. Future research should explore whether the premium reflects compensation for systematic risk, behavioral biases, or institutional frictions. Second, our analysis focuses primarily on U.S. equity markets; examining the international robustness of IFC across different market structures and regulatory environments would enhance our understanding of its generalizability. Third, investigating the time-varying nature of the IFC premium and its relationship to macroeconomic conditions could provide insights into when the signal is most effective.

Future research should also examine the interaction between IFC and corporate governance mechanisms, as financing conservatism may reflect managerial quality or agency considerations. Additionally, exploring the signal's performance across different firm characteristics such as size, age, and financial constraints could reveal important heterogeneity in its effectiveness. Finally, investigating whether machine learning techniques can enhance the signal's construction or identify nonlinear relationships would align with recent advances in empirical asset pricing methodology.

In conclusion, this paper establishes Investment Financing Conservatism as a robust and economically significant predictor of stock returns that adds value beyond existing factor models and related anomalies. While questions remain about its theoretical foundations and boundary conditions, the evidence strongly supports its inclusion in the constellation of documented return predictors and highlights the continued importance of corporate financing decisions in asset pricing.

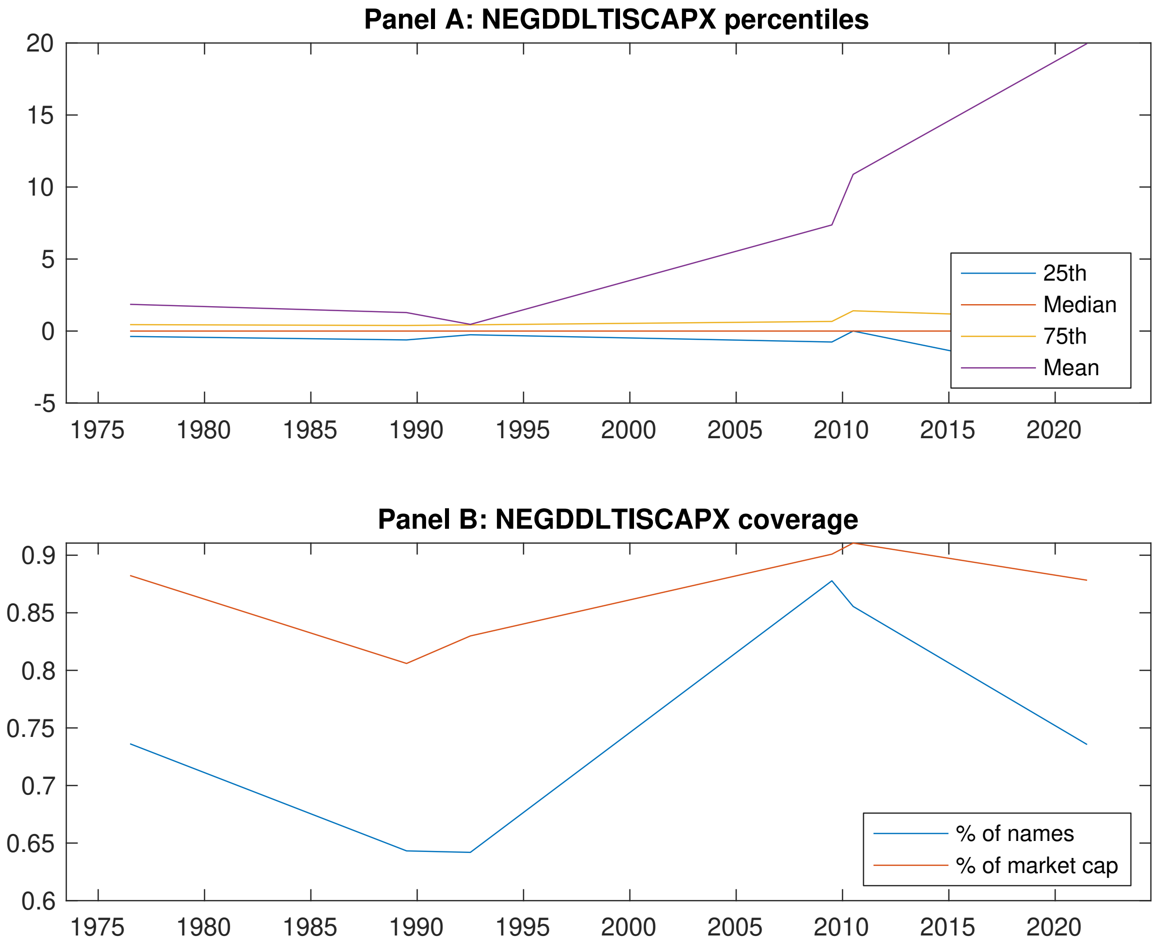


Figure 1: Times series of IFC percentiles and coverage.
This figure plots descriptive statistics for IFC. Panel A shows cross-sectional percentiles of IFC over the sample. Panel B plots the monthly coverage of IFC relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on IFC. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 197306 to 202406.

Panel A: Excess returns and alphas on IFC-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.55	0.67	0.65	0.72	0.84	0.29
	[2.56]	[3.76]	[3.26]	[4.05]	[4.19]	[4.26]
α_{CAPM}	-0.18	0.07	-0.02	0.12	0.16	0.34
	[-3.44]	[1.35]	[-0.33]	[2.35]	[3.00]	[5.05]
α_{FF3}	-0.20	0.04	0.05	0.12	0.13	0.34
	[-3.91]	[0.97]	[0.93]	[2.38]	[2.57]	[4.95]
α_{FF4}	-0.17	0.04	0.10	0.07	0.13	0.30
	[-3.21]	[0.91]	[1.86]	[1.44]	[2.40]	[4.30]
α_{FF5}	-0.16	-0.03	0.10	0.03	0.08	0.24
	[-3.10]	[-0.77]	[1.75]	[0.65]	[1.48]	[3.54]
α_{FF6}	-0.14	-0.03	0.13	0.00	0.08	0.22
	[-2.65]	[-0.68]	[2.41]	[0.06]	[1.47]	[3.18]
Panel B: Fama and French (2018) 6-factor model loadings for IFC-sorted portfolios						
β_{MKT}	1.08	0.98	0.97	0.96	1.05	-0.03
	[89.35]	[95.69]	[75.06]	[83.22]	[84.68]	[-1.99]
β_{SMB}	0.11	-0.12	-0.02	-0.02	0.11	0.00
	[6.08]	[-7.63]	[-0.83]	[-1.12]	[6.04]	[0.09]
β_{HML}	0.08	0.06	-0.17	-0.08	-0.02	-0.10
	[3.42]	[2.92]	[-6.82]	[-3.57]	[-0.83]	[-3.28]
β_{RMW}	0.01	0.14	-0.04	0.08	0.08	0.07
	[0.36]	[7.09]	[-1.40]	[3.53]	[3.09]	[2.15]
β_{CMA}	-0.16	0.10	-0.10	0.22	0.13	0.29
	[-4.48]	[3.20]	[-2.76]	[6.55]	[3.62]	[6.28]
β_{UMD}	-0.04	-0.01	-0.06	0.05	-0.00	0.04
	[-3.03]	[-0.59]	[-4.54]	[4.10]	[-0.02]	[2.31]
Panel C: Average number of firms (n) and market capitalization (me)						
n	674	504	1021	562	636	
me ($\$10^6$)	1502	2808	2233	2766	1568	

Table 2: Robustness to sorting methodology & trading costs
This table evaluates the robustness of the choices made in the IFC strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 197306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.29	0.34	0.34	0.30	0.24	0.22
			[4.26]	[5.05]	[4.95]	[4.30]	[3.54]	[3.18]
Quintile	NYSE	EW	0.19	0.21	0.20	0.19	0.18	0.18
			[4.18]	[4.65]	[4.38]	[4.13]	[4.01]	[3.91]
Quintile	Name	VW	0.26	0.31	0.33	0.29	0.26	0.24
			[3.58]	[4.26]	[4.46]	[3.88]	[3.63]	[3.31]
Quintile	Cap	VW	0.31	0.35	0.36	0.31	0.28	0.25
			[4.74]	[5.46]	[5.68]	[4.82]	[4.41]	[3.87]
Decile	NYSE	VW	0.32	0.37	0.37	0.36	0.24	0.25
			[3.64]	[4.16]	[4.21]	[4.04]	[2.78]	[2.85]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r^e_{net}	α^*_{CAPM}	α^*_{FF3}	α^*_{FF4}	α^*_{FF5}	α^*_{FF6}
Quintile	NYSE	VW	0.23	0.28	0.27	0.25	0.19	0.17
			[3.36]	[4.07]	[3.93]	[3.61]	[2.81]	[2.57]
Quintile	NYSE	EW	-0.08					
			[-1.37]					
Quintile	Name	VW	0.20	0.24	0.25	0.23	0.21	0.19
			[2.74]	[3.28]	[3.44]	[3.14]	[2.80]	[2.57]
Quintile	Cap	VW	0.25	0.29	0.30	0.27	0.23	0.20
			[3.92]	[4.48]	[4.65]	[4.22]	[3.59]	[3.28]
Decile	NYSE	VW	0.25	0.29	0.29	0.29	0.18	0.18
			[2.79]	[3.21]	[3.23]	[3.17]	[2.06]	[2.02]

Table 3: Conditional sort on size and IFC

This table presents results for conditional double sorts on size and IFC. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on IFC. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high IFC and short stocks with low IFC .Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 197306 to 202406.

Panel A: portfolio average returns and time-series regression results

		IFC Quintiles					IFC Strategies					
		(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Size quintiles	(1)	0.72	0.90	0.91	0.95	0.77	0.05	0.07	0.05	0.05	0.08	0.08
		[2.65]	[3.26]	[3.25]	[3.32]	[2.84]	[0.70]	[0.90]	[0.70]	[0.65]	[1.00]	[0.97]
	(2)	0.81	0.81	0.87	0.91	0.81	0.01	0.04	0.00	0.01	-0.04	-0.03
		[3.06]	[3.20]	[3.41]	[3.64]	[3.24]	[0.08]	[0.52]	[0.04]	[0.16]	[-0.50]	[-0.36]
	(3)	0.74	0.85	0.80	0.79	0.93	0.19	0.23	0.22	0.18	0.21	0.18
		[3.01]	[3.76]	[3.30]	[3.48]	[3.98]	[2.57]	[3.17]	[3.03]	[2.37]	[2.85]	[2.38]
	(4)	0.74	0.80	0.82	0.78	0.88	0.14	0.16	0.14	0.12	0.09	0.09
		[3.34]	[3.76]	[3.71]	[3.63]	[4.07]	[1.88]	[2.19]	[1.93]	[1.68]	[1.26]	[1.15]
	(5)	0.45	0.63	0.59	0.62	0.84	0.39	0.42	0.45	0.37	0.36	0.31
		[2.20]	[3.55]	[3.05]	[3.41]	[4.34]	[4.43]	[4.78]	[5.02]	[4.21]	[4.01]	[3.49]

Panel B: Portfolio average number of firms and market capitalization

		IFC Quintiles Average n					IFC Quintiles Average market capitalization ($\$10^6$)				
		(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)
Size quintiles	(1)	380	381	381	381	377	35	34	31	33	34
	(2)	105	105	105	105	105	60	60	59	60	59
	(3)	75	75	75	75	75	106	106	103	104	106
	(4)	63	63	63	63	62	226	234	225	230	224
	(5)	57	57	57	57	57	1328	2082	1748	2145	1446

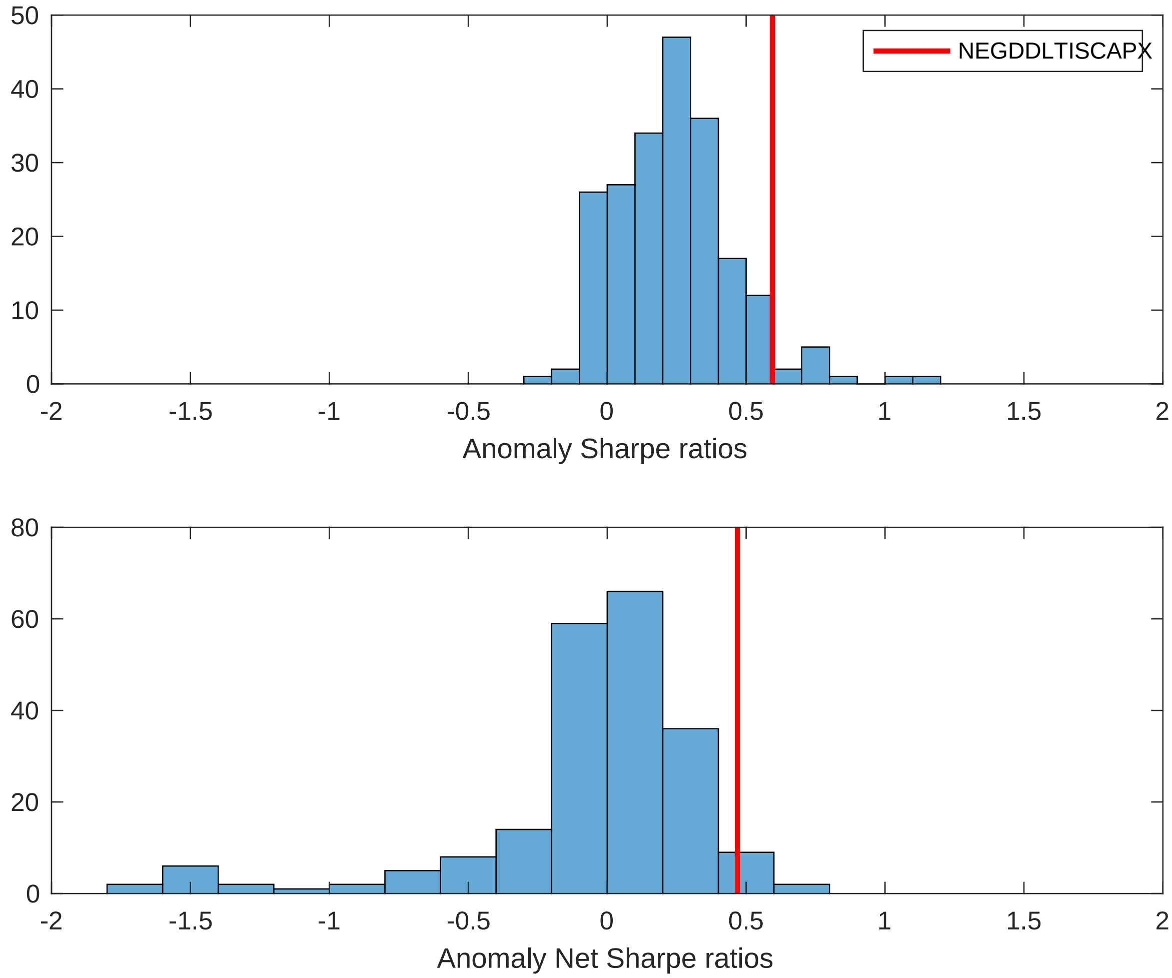


Figure 2: Distribution of Sharpe ratios.
This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the IFC with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

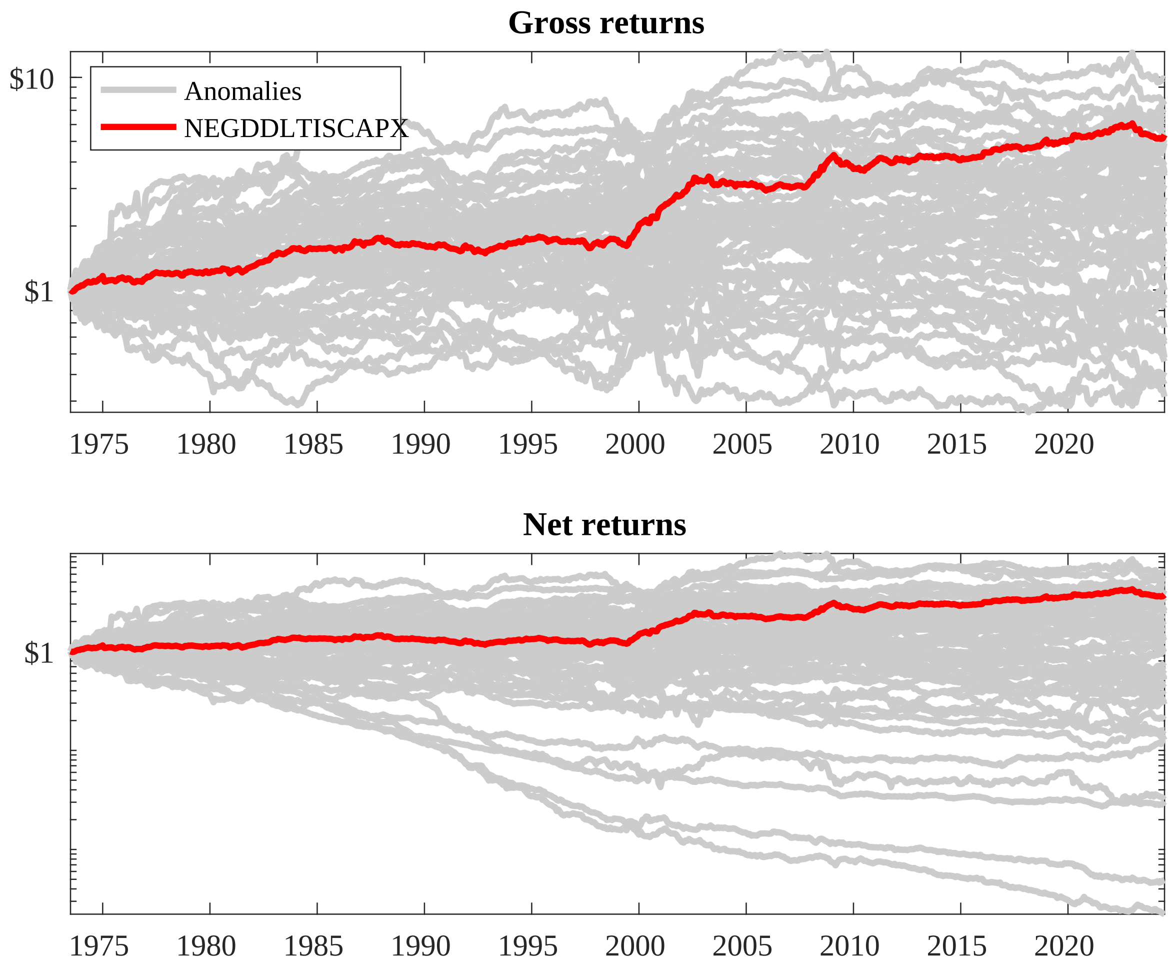
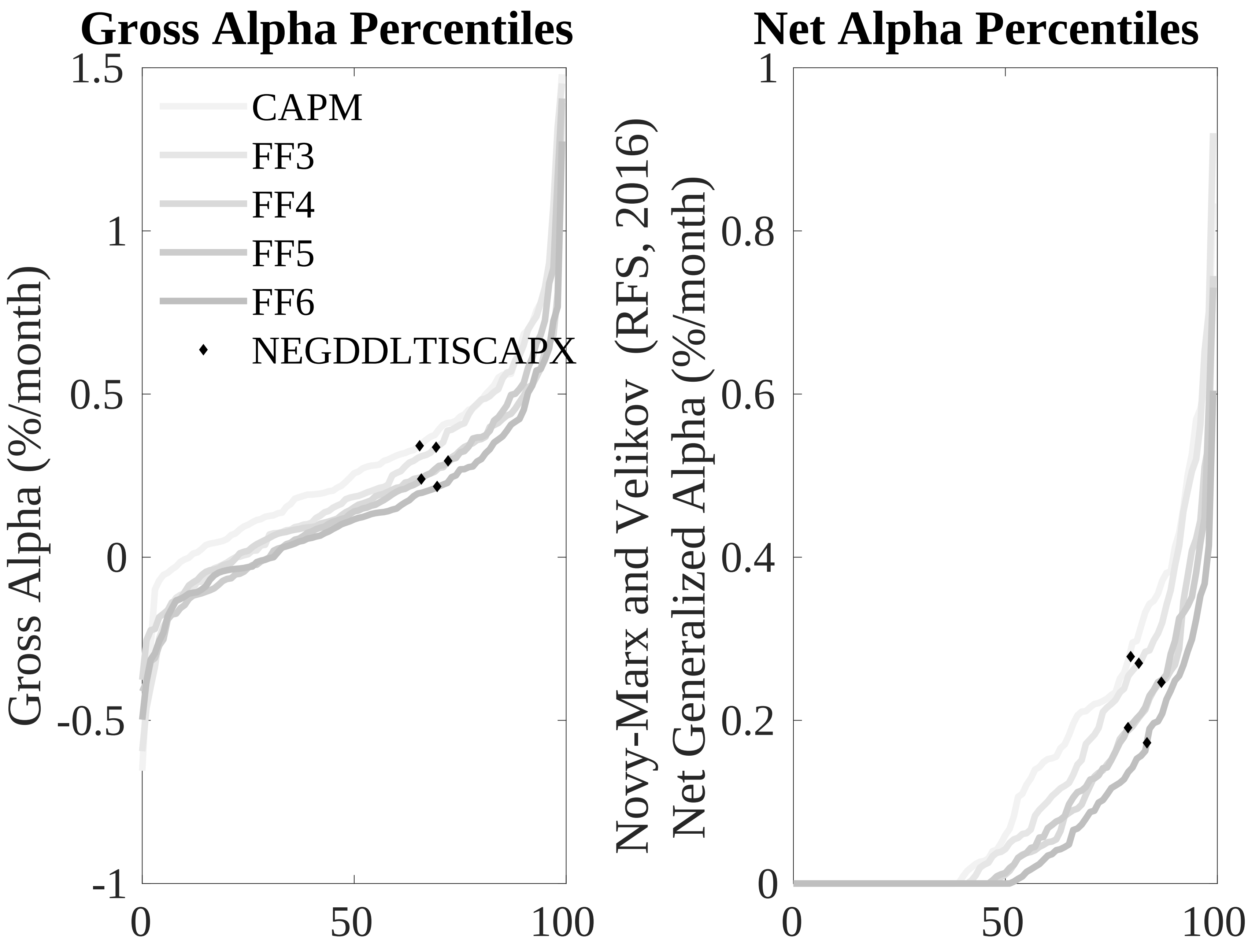
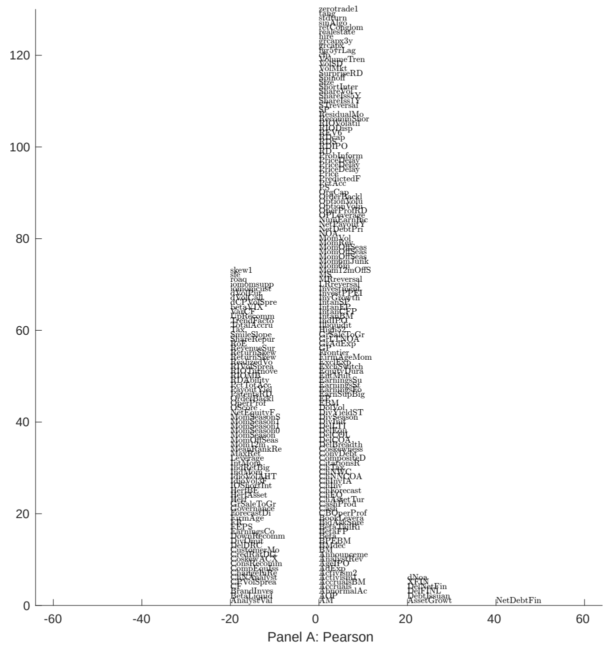


Figure 3: Dollar invested.
This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the IFC trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strtaegy returns.


Gross Alpha Percentiles
Net Alpha Percentiles
1.5
1
0.5
0
-0.5
-1
0.8
0.6
0.4
0.2
0
50
100
CAPM
FF3
FF4
FF5
FF6
NEGDDLTISCAPX
Gross Alpha (%/month)
Novy-Marx and Velikov (RFS, 2016)
Net Generalized Alpha (%/month)
Anomaly percentile rank
Anomaly percentile rank



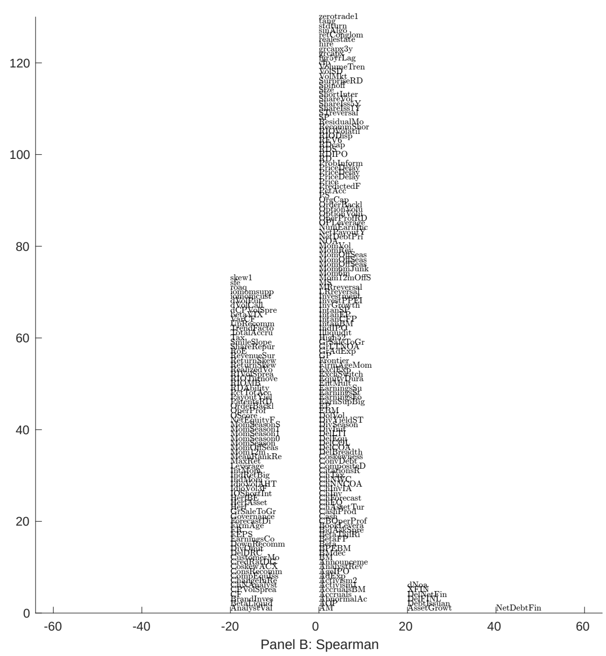


Figure 5: Distribution of correlations.
This figure plots a name histogram of correlations of 210 filtered anomaly signals with IFC. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

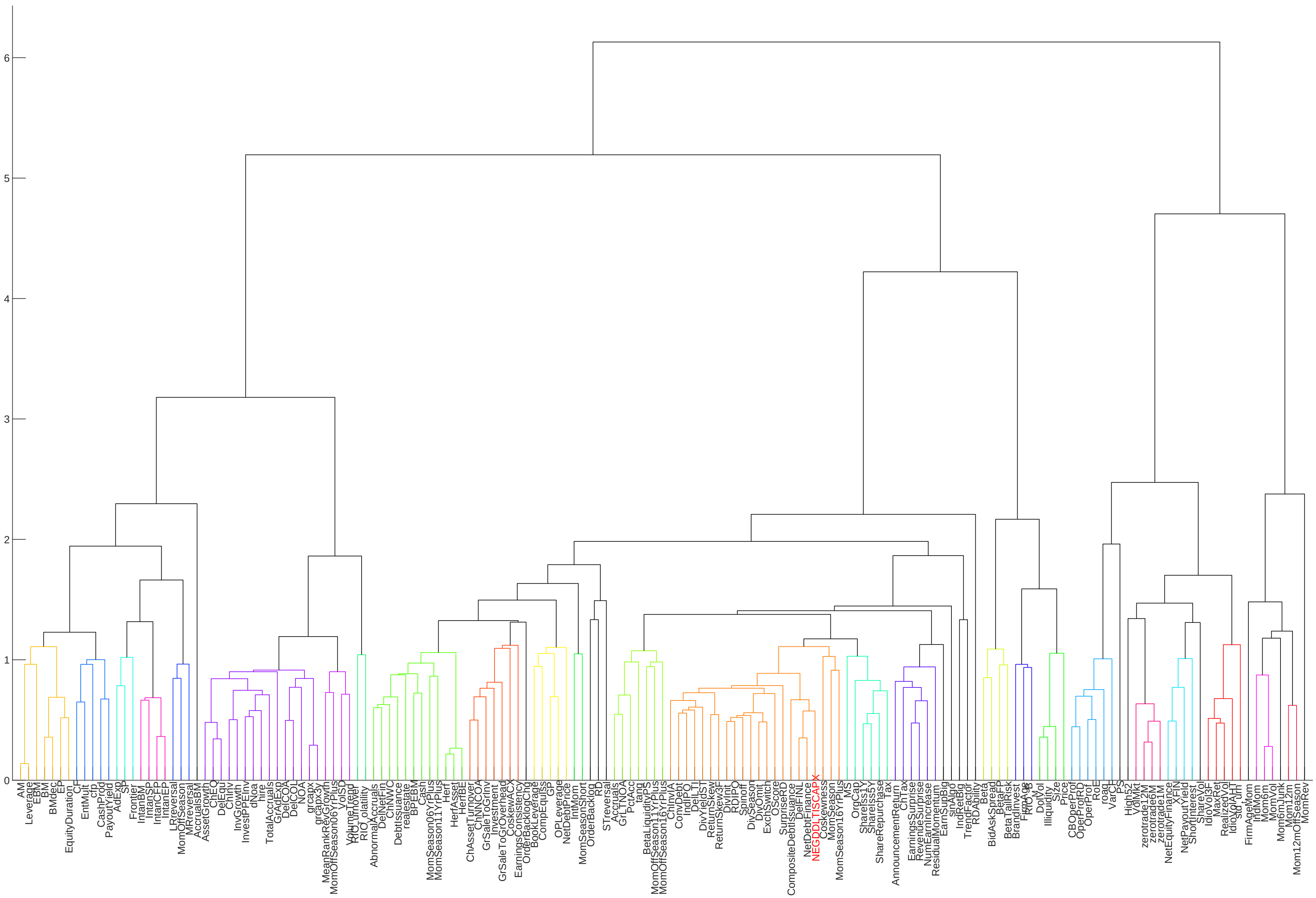


Figure 6: Agglomerative hierarchical cluster plot
This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

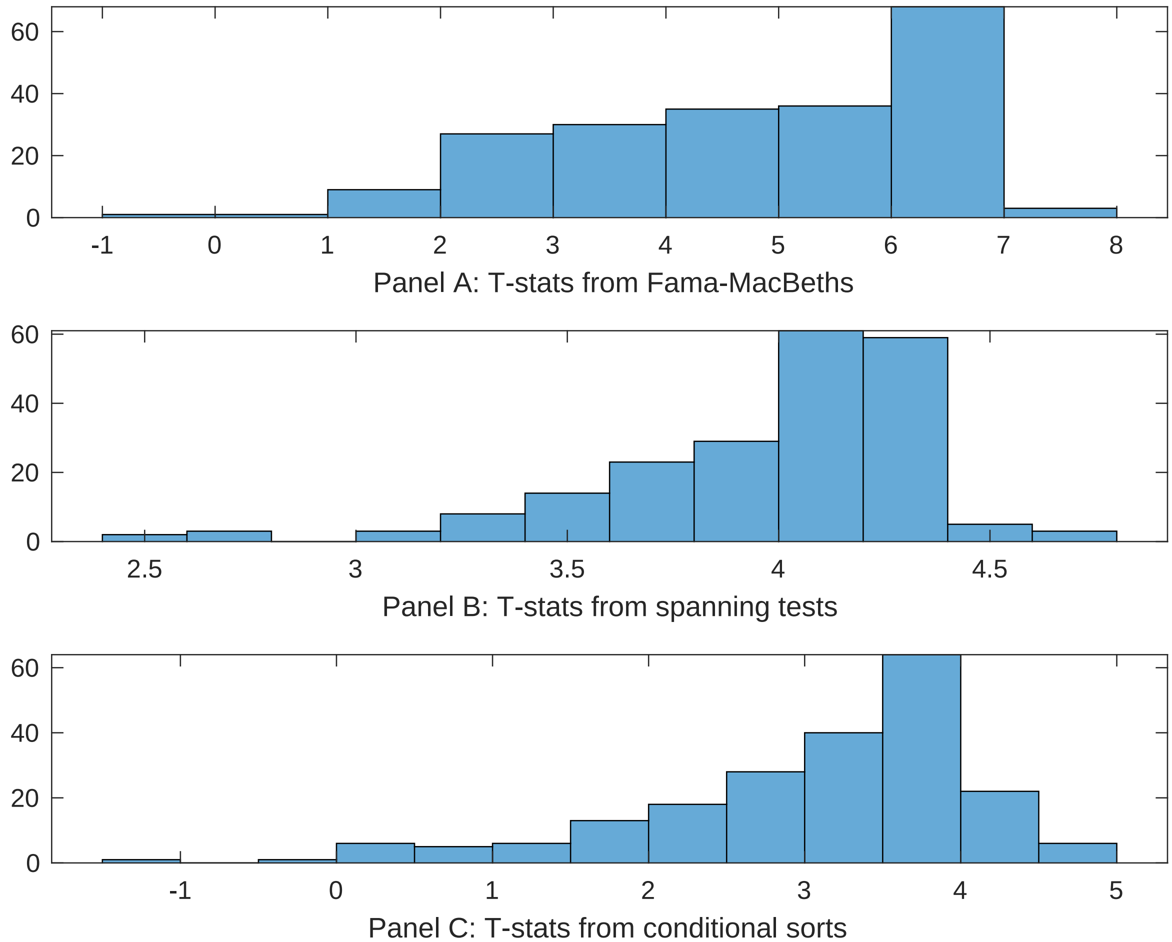


Figure 7: Distribution of t-stats on conditioning strategies
This figure plots histograms of t-statistics for predictability tests of IFC conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{IFC} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{IFC} IFC_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{IFC,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on IFC. Stocks are finally grouped into five IFC portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted IFC trading strategies conditioned on each of the 210 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies
This table presents Fama-MacBeth results of returns on IFC. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{IFC} IFC_{i,t} + \sum_{k=1}^{s} ix\beta_{X_k} X_{i,t}^{k} + \epsilon_{i,t}$. The six most closely related anomalies, X, are Change in financial liabilities, Net external financing, Net debt financing, Asset growth, change in net operating assets, Employment growth. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 197306 to 202406.

Intercept	0.13	0.14	0.13	0.14	0.14	0.13	0.15
	[5.38]	[5.69]	[5.34]	[5.81]	[5.68]	[5.35]	[5.96]
IFC	0.51	0.80	0.61	0.54	0.45	0.13	0.19
	[2.10]	[3.11]	[2.32]	[2.05]	[1.76]	[5.28]	[0.71]
Anomaly 1	0.17						-0.54
	[9.10]						[-1.32]
Anomaly 2		0.20					0.80
		[6.43]					[1.53]
Anomaly 3			0.21				0.60
			[9.11]				[1.06]
Anomaly 4				0.11			0.49
				[8.61]			[2.64]
Anomaly 5					0.14		0.73
					[9.67]		[4.55]
Anomaly 6						0.95	0.12
						[6.20]	[0.98]
# months	612	612	612	612	612	612	612
$\bar{R}^2$(%)	0	1	0	0	0	0	0

Table 5: Spanning tests controlling for most closely related anomalies
This table presents spanning tests results of regressing returns to the IFC trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{IFC} = \alpha + \sum_{k=1}^{6} \beta_{X_k} r_t^{X_k} + \sum_{j=1}^{6} \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor. The six most closely related anomalies, X, are Change in financial liabilities, Net external financing, Net debt financing, Asset growth, change in net operating assets, Employment growth. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 197306 to 202406.

Intercept	0.19	0.20	0.20	0.21	0.21	0.22	0.20
	[2.86]	[2.89]	[2.93]	[3.05]	[3.01]	[3.18]	[2.90]
Anomaly 1	14.96						11.09
	[3.91]						[2.10]
Anomaly 2		10.79					7.41
		[3.18]					[2.02]
Anomaly 3			15.30				7.07
			[4.15]				[1.40]
Anomaly 4				8.20			6.10
				[2.08]			[1.41]
Anomaly 5					1.70		-7.28
					[0.43]		[-1.66]
Anomaly 6						3.45	2.09
						[0.94]	[0.54]
mkt	-2.57	-1.43	-2.88	-2.91	-2.91	-2.82	-1.59
	[-1.65]	[-0.88]	[-1.86]	[-1.86]	[-1.85]	[-1.79]	[-0.97]
smb	-0.73	4.27	-0.55	0.21	0.84	1.04	0.99
	[-0.31]	[1.64]	[-0.23]	[0.09]	[0.35]	[0.43]	[0.36]
hml	-8.39	-8.18	-8.76	-9.77	-9.51	-10.04	-7.62
	[-2.79]	[-2.70]	[-2.93]	[-3.23]	[-3.12]	[-3.22]	[-2.45]
rmw	5.31	0.21	4.97	6.68	6.68	6.81	0.45
	[1.73]	[0.06]	[1.62]	[2.18]	[2.17]	[2.21]	[0.12]
cma	23.78	21.75	24.82	18.86	27.52	25.70	14.55
	[5.04]	[4.29]	[5.35]	[2.85]	[5.08]	[4.55]	[2.08]
umd	2.47	3.69	2.78	3.93	3.71	3.54	2.52
	[1.56]	[2.38]	[1.78]	[2.52]	[2.37]	[2.24]	[1.59]
# months	612	612	612	612	612	612	612
$\bar{R}^2$(%)	13	13	14	12	11	11	15

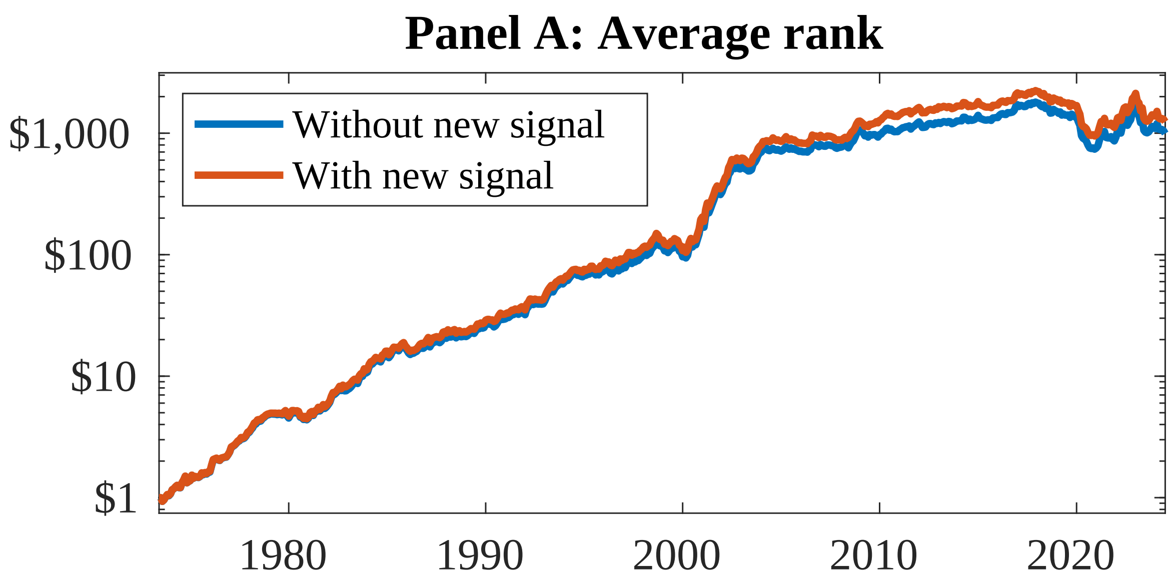


Figure 8: Combination strategy performance
This figure plots the growth of a $1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). In all panels, the blue solid lines indicate combination trading strategies that utilize 155 anomalies. The red solid lines indicate combination trading strategies that utilize the 155 anomalies as well as IFC. Panel A shows results using "Average rank" as the combination method. See Section 7 for details on the combination methods.

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